

Overview of Virtualization and Benefits-18

Virtualization is a technology that enables the creation of virtual environments from a single physical machine, allowing for more efficient use of resources by distributing them across computing environments.

Using software, virtualization creates an abstraction layer over computer hardware, dividing a single system's components such as processors, memory, networks and storage into multiple virtual machines (VM).

Each VM runs its own operating systems and behaves like a separate physical computer, despite sharing the same underlying hardware.



It allows cloud service providers (CSPs) such as IBM Cloud®, Microsoft Azure, Google Cloud and Amazon Web Services (AWS), to optimally utilize their IT infrastructure to deliver scalable resources. For businesses, this means they only purchase the computing resources they need and then scale them cost-effectively as their workloads grow, maximizing their investment. **Types of Virtualization**

- **Desktop virtualization**
Runs user desktops on virtual machines in a data center. Enables remote access, central management, and secure work environments.
- **Network virtualization**
Abstracts physical networking hardware into logical networks. Supports better segmentation, security, and flexible management.
- **Storage virtualization**
Combines multiple storage devices into a single virtual pool. Simplifies management and improves utilization of storage resources.
- **Data virtualization**

Provides a single access layer to data from different sources without needing to move or copy it. Useful for analytics and integration.

- **Application virtualization**

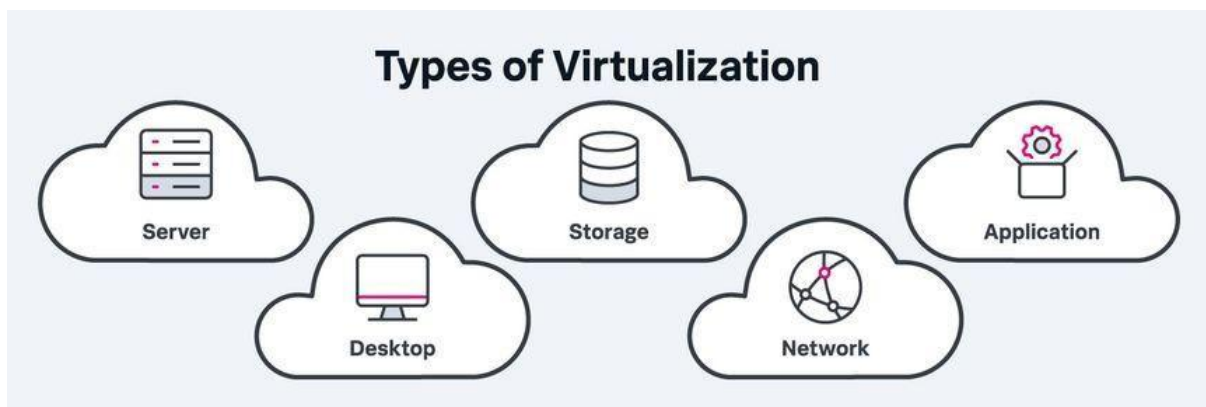
Runs applications in isolated virtual environments. Reduces compatibility issues and simplifies deployment across systems.

- **Data center virtualization**

Abstracts the entire data center infrastructure, including servers, storage, and networks. Creates software-defined data centers (SDDC).

- **Cloud virtualization**

Provides virtualized resources such as servers, storage, and applications through cloud platforms. Key to IaaS, PaaS, and SaaS models.



Its key benefits

- **Better use of hardware**

Physical servers often idle. Virtualization lets each server run many virtual machines, increasing utilization.

- **Cost savings**

You need fewer physical machines, which lowers expenses on hardware, energy, space, and maintenance.

- **Scalability and flexibility**

New virtual machines spin up quickly. You can adjust computing resources up or down on the fly.

- **Stronger isolation and security**

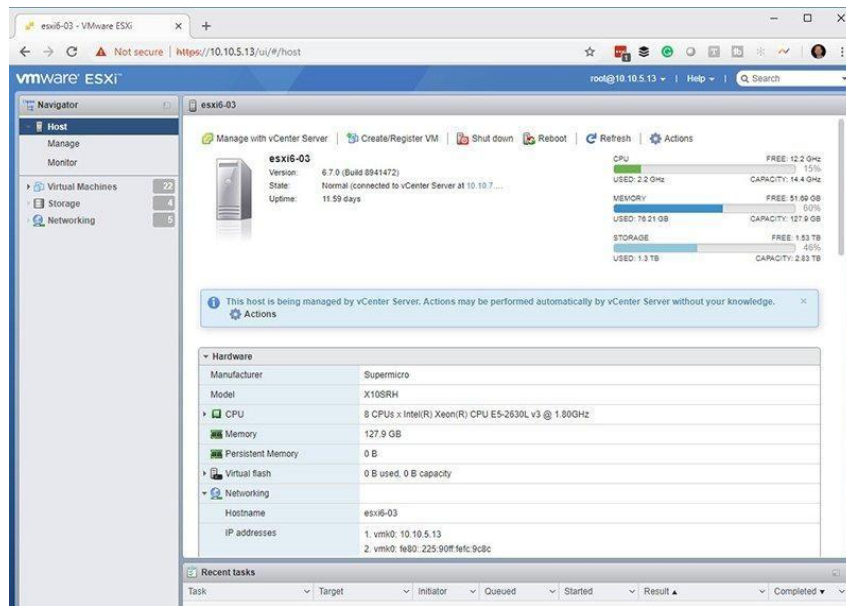
VMs run separately. If one fails or is compromised, others stay safe. Isolation helps contain threats.

Benefits of Virtualisation



- **Better uptime and resilience**
Virtualization supports live migration, automated failover, snapshots, and backups. That enables fast recovery from failures.
- **Simplified management**
IT can manage VMs from centralized dashboards. Tasks like deployments, patches, and scaling become automated.
- **Disaster recovery (DR)**
Virtualization optimizes disaster recovery by enabling quick restoration of services with minimal downtime. Since virtual machines can be easily moved, replicated or backed up

vSphere

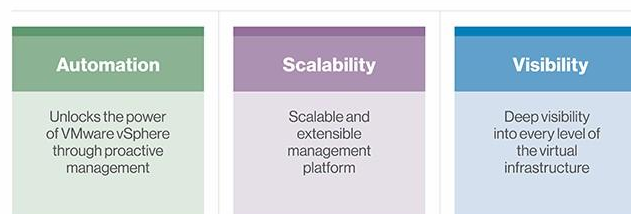


- VMware's virtualization platform that provides the foundation for creating and running virtual machines on physical servers. Includes the ESXi hypervisor, which abstracts hardware resources and allows multiple VMs to run on the same server.

vCenter

- Centralized management tool for vSphere environments. Enables administrators to manage multiple ESXi hosts and VMs from one console.

VMware vCenter Server



Main components

- Physical machine (Host machine) - provides CPU, memory, storage and network resources for the virtual machines.
- Virtual machine (VMs) - virtual environment that simulates a physical computer in software form.
- Hypervisor - it is the software layer that coordinates VMs.